

The volume bound charge density is given by $\rho_b = -\nabla \cdot \mathbf{P}$. With $\mathbf{P} = k\mathbf{r}$, we have

$$-\nabla \cdot \mathbf{P} = -k(\partial_x x + \partial_y y + \partial_z z) = -3k.$$

The surface bound charge density is given by $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$. Since the cube is at the origin, the faces are at $x = \pm a/2$, or the same for y or z . The computation for σ_b is the same on all faces with a coordinate at $a/2$, and at $-a/2$, so we show two computations. Take $x = a/2$. Then the normal vector is $\hat{\mathbf{x}}$, so

$$\mathbf{P} \cdot \hat{\mathbf{x}} = (ka/2, ky, kz) \cdot (1, 0, 0) = \frac{ka}{2}.$$

Now take $x = -a/2$. The normal vector is $-\hat{\mathbf{x}}$, so

$$\mathbf{P} \cdot \hat{\mathbf{x}} = (-ka/2, ky, kz) \cdot (-1, 0, 0) = \frac{ka}{2}.$$

Therefore, $\sigma_b = ka/2$ on any face.

We can now directly compute the bound charges. Since the volume is a^3 , the volume bound charge is

$$a^3 Q_v = \int_V \rho_b dV' = (\rho_b \cdot V) = -3ka^3.$$

The surface bound charge density is the same on all faces, so we can compute it for a single face S and multiply by 6. This gives

$$Q_s = 6 \int_S \sigma_b dS = 6 \left(\frac{ka}{2} \cdot S \right) = 3ka^3.$$

They indeed cancel.